

Z₂-Equivariant World Models for Imagination Training in a First-Person Shooter: Reward-Head Stability without Return Gains

Anonymous^a

^a*Anonymous Institution (double-blind review)*

Abstract

First-person shooter (FPS) games stress deep reinforcement learning (RL): visual observations are high-dimensional and sample complexity is prohibitive. World models cut real-environment interaction by training the policy inside a learned dynamics model, but most implementations ignore the left-right mirror symmetry that FPS environments possess almost by construction. We study *EqWM*, a compact world model and PPO policy that enforce horizontal-flip ($\mathbb{Z}_2$) equivariance by group averaging across encoder, recurrent dynamics, decoder, and reward head, on the ViZDoom `health_gathering` benchmark [1]. We report findings without inflation; the honest headline is stability rather than returns. The equivariant reward head is markedly more stable across seeds, with reward-prediction MAE 0.104 ± 0.025 versus 0.223 ± 0.099 , but a constant-predictor analysis shows the reward is near-degenerate and both heads barely exceed predicting its mean: the benefit is cross-seed stability of a task-relevant but not well-learned head, not evidence of reward understanding. Single-frame reconstruction is statistically indistinguishable and marginally favors the non-equivariant model. Imagination training reaches, after about 15k real steps, returns comparable to pure PPO at 20k–40k steps, an order-of- $2\times$ sample-efficiency gain that we mark as preliminary, with a measured step ratio of $1.3\text{--}2.7\times$ and flat noisy curves. Under this small-model budget, equivariant and non-equivariant imagination variants are not distinguishable, and a four-cell ablation pooled over nine same-machine seeds finds no significant return gain, with the fully equivariant configuration numerically highest but far from significance. A bootstrap estimate of the effective-sample multiplier is non-monotone and cross-machine inconsistent. The theory is presented as upper bounds and

an explicitly qualitative scaling law, with every experimental support, weak support, or contradiction marked.

Keywords: Equivariant neural networks, World models, Imagination training, Reinforcement learning, First-person shooter, PAC-Bayes

1. Introduction

Deep reinforcement learning (RL) has reached superhuman levels in many games, but training an agent directly from raw pixels in first-person shooters (FPS) still demands millions of environment interactions [2, 1], which is impractical when environment access is expensive. *World models* address this by learning a predictive model of the environment and optimizing the policy entirely inside it [3]. The Dreamer family operationalizes this idea with a recurrent state-space model [4, 5, 6] and has recently scaled to long-horizon tasks inside learned models [7]; model-based planning with a learned dynamics model is the core of MuZero [8], and model-based policy optimization exploits such models for data efficiency [9].

FPS environments, however, possess a near-perfect structural symmetry that none of these general-purpose architectures exploit: the rendered scene and its dynamics are (approximately) invariant under a horizontal flip, and a flipped camera view should induce flipped actions ($\text{TURN_LEFT} \leftrightarrow \text{TURN_RIGHT}$, MOVE_FORWARD unchanged). Group-equivariant networks [10] bake such symmetries into the architecture and, in principle, use each observation several times at no extra data cost. This is precisely the data-starved regime in which world models operate.

We ask a deliberately narrow question: *on a small FPS benchmark, does $\mathbb{Z}_2$ equivariance in the world model and policy improve sample efficiency or final return, and how should that be measured honestly?* Our system, *EqWM*, enforces hard equivariance by group averaging [10] at every convolutional layer of the encoder, recurrent dynamics, decoder, and policy, with an invariant reward head and an antisymmetric left/right action head. Our contributions are:

1. **Implementation and controlled comparison on ViZDoom.** We build an equivariant ConvGRU world model and equivariant PPO policy for `health_gathering` and compare them against matched non-equivariant baselines, reporting per-seed, aggregated, and pooled-seed statistics (Section 6).

2. **PAC-Bayes-style analysis of imagination training.** We state three results: a world-model prediction bound that depends on an effective sample count N_{eff} (Theorem 1), an imagination-training generalization bound that separates world-model bias from estimation error (Theorem 2), and a scaling law for the optimal real-to-imagination ratio (Corollary 3); proofs are given in Appendix Appendix A, and a bootstrap estimator with an explicit bias-variance caveat (Proposition 1) is provided.
3. **Honest empirical verdict.** The clearest measured benefit of equivariance is *cross-seed stability of the reward head*, which a constant-predictor analysis shows to be stability of a near-degenerate predictor rather than evidence of reward understanding; imagination training gives an order-of- $2\times$ sample-efficiency gain over pure PPO that is preliminary; a four-cell ablation pooled over nine same-machine seeds shows no significant return gain from any equivariant component; and the bootstrap estimate of the effective-sample multiplier is not robust. We treat the weak and negative results as limitations and open problems rather than burying them.

The remainder of the paper is organized as follows. Section 2 reviews related work. Section 3 fixes notation. Section 4 details the architecture. Section 5 states the theory. Section 6 reports experiments, and Section 7 concludes with limitations.

2. Related Work

World models and imagination training. Learning a predictive model for RL dates back to Ha and Schmidhuber [3]. The Dreamer series [4, 5] operationalized latent imagination with a recurrent state-space model, and DreamerV3 [6] showed a single model can solve many domains; the most recent Dreamer4 [7] scales agent training inside a fast world model in Minecraft. MuZero [8] pairs learned dynamics with planning, and MBPO [9] uses an ensemble of models to cut real samples. Concurrent with these, EMERALD [11] improves world modeling with masked latent transformers and strong sample efficiency on Crafter. Our work follows the Dreamer-style “train PPO inside a learned latent model” loop but restricts the model class to a group-equivariant one.

Equivariant reinforcement learning. Group-equivariant convolutions were introduced by Cohen and Welling [10]. In RL, MDP homomorphic networks [12] and group equivariant deep RL [13] encode state/action symmetries,

while $\text{SO}(2)$ -equivariant policies [14] and group-invariant representation structuring [15] exploit continuous rotation symmetries. The closest lines to ours are equivariant model-based RL: Park et al. learn symmetric embeddings for equivariant world models [16], Deac et al. make MuZero equivariant [17], and EDGI builds an $\text{SE}(3)$ -equivariant diffusion planner for embodied agents [18]. Multi-group equivariant augmentation [19] studies several symmetry groups in manipulation. Unlike these works, which target control, board games, or 3D/ $\text{SE}(3)$ embodied settings, we target 2D FPS pixels with the discrete $\mathbb{Z}_2$ group and integrate equivariance into *both* the world model and the imagination-training loop, while reporting the small-benchmark result transparently.

PAC-Bayes and generalization in RL. PAC-Bayes bounds [20] relate a posterior’s risk to its empirical risk through a KL complexity term. Recent PAC-Bayesian RL [21] trains generalizable on-policy policies. We adapt the decomposition to the mixed real/imagination data regime and use the bootstrap [22] to estimate the effective sample multiplier empirically.

FPS reinforcement learning. ViZDoom [1] provides a Doom-based visual-RL benchmark; early agents trained directly with model-free RL [2]. To our knowledge, prior work has not combined an equivariant world model with imagination training on FPS pixels.

3. Preliminaries

MDP. We model the environment as an MDP $\langle \mathcal{S}, \mathcal{A}, P, r, \gamma \rangle$ [23], where $\mathcal{S}$ is the RGB frame space, $\mathcal{A} = \{\text{TURN-LEFT}, \text{TURN-RIGHT}, \text{FORWARD}\}$ is a three-action discrete set, P the transition kernel, r the reward, and $\gamma = 0.99$ the discount. The objective is the expected discounted return $J(\pi) = \mathbb{E}_\pi[\sum_t \gamma^t r_t]$.

Group action and equivariance. We take $G = \mathbb{Z}_2 = \{e, h\}$, where h is a horizontal flip. The group acts on observations by $h \cdot x = \text{flip}_W(x)$ and on actions by the permutation $h \cdot \text{TURN-LEFT} = \text{TURN-RIGHT}$, $h \cdot \text{TURN-RIGHT} = \text{TURN-LEFT}$, $h \cdot \text{FORWARD} = \text{FORWARD}$. A map f is *equivariant* if $f(gx) = gf(x)$ and *invariant* if $f(gx) = f(x)$.

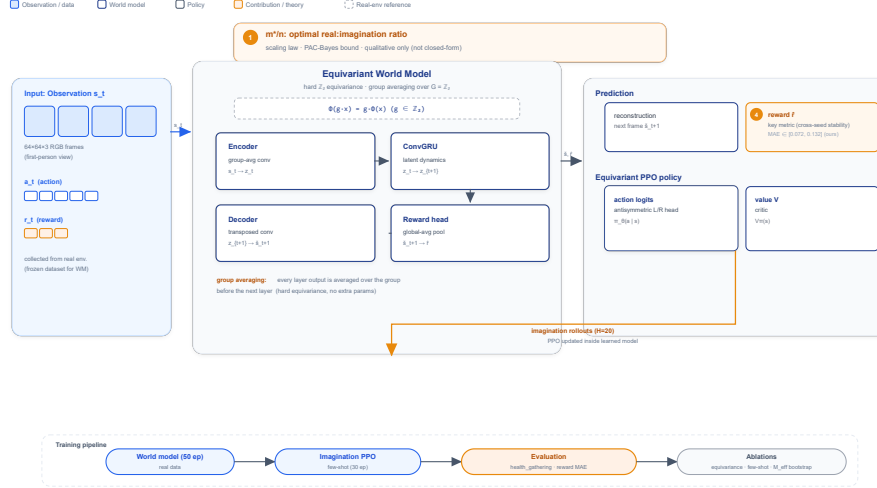


Figure 1: Overview of the equivariant world model and imagination training. The world model enforces hard $\mathbb{Z}_2$ equivariance by group averaging across an Encoder, a ConvGRU dynamics cell, a Decoder, and a reward head; it predicts both frame reconstruction and a scalar reward. The equivariant PPO policy is updated inside the learned model on imagination rollouts, and the analysis bounds the optimal real:imagination ratio m^*/n only qualitatively.

World models and imagination.. A world model $\hat{M} = \langle \hat{P}, \hat{r} \rangle$ approximates the environment. Given $\hat{M}$, imagination training starts a policy rollout from a real observation and predicts subsequent latent states and rewards with $\hat{M}$, then updates the policy (here PPO [24]) on these imagined trajectories [6].

PAC-Bayes.. With posterior ρ , prior π , and n i.i.d. samples, a canonical bound is $R(\rho) \leq R_S(\rho) + \sqrt{(\text{KL}(\rho||\pi) + \log(2\sqrt{n}/\delta))/(2n)}$.

4. Method

4.1. Equivariant $\mathbb{Z}_2$ world model

The world model $\hat{M}_\phi$ has four parts (Figure 1). *Encoder* E_ϕ : three strided convolutions map a $64 \times 64 \times 3$ frame to an $8 \times 8 \times z$ latent map ($z = 32$, base channels 16). Every convolution is a group-averaging layer [10],

$$E_\phi(x) = \frac{1}{2} [\text{Conv}(x) + \text{flip}_W(\text{Conv}(\text{flip}_W(x)))], \quad (1)$$

which shares one kernel and satisfies $E_\phi(\text{flip}_W x) = \text{flip}_W E_\phi(x)$ up to numerical precision. *Dynamics*: an equivariant ConvGRU cell updates the latent

map from the previous map and a spatially broadcast one-hot action; gates use the same group-averaging convolution. *Decoder*: three equivariant transposed convolutions reconstruct the frame with a sigmoid output. *Reward head*: a global-average-pooling (invariant) layer followed by an MLP predicts a scalar reward from the latent map. The non-equivariant baseline uses identical widths and layer counts but ordinary convolutions, a flattened encoder for its heads, and no group averaging.

The training loss is reconstruction MSE plus reward MSE plus latent-dynamics MSE, and (only for the equivariant model) a soft consistency penalty on the encoder and dynamics under flip.

4.2. Equivariant PPO policy

The policy uses the same equivariant encoder. Invariant features ($\text{gap}(h)$) drive the forward action and the (invariant) value head; an antisymmetric feature $a = \overline{h_{\text{left}}} - \overline{h_{\text{right}}}$ distinguishes left and right: $\ell_{\text{left}} = s + a$, $\ell_{\text{right}} = s - a$, $\ell_{\text{forward}} = s'$. This enforces $\pi(h \cdot x) = h \cdot \pi(x)$.

4.3. Imagination pipeline

We collect $n = 10\text{k}$ random real transitions, train $\hat{M}_\phi$ for 30 epochs, then iterate: collect 1,000 new on-policy real transitions, fine-tune $\hat{M}_\phi$ for 5 epochs, imagine short rollouts of horizon $H = 20$ under PPO [24] (200 rollouts per outer step), and update on the imagined data. This gives roughly 4,000 imagined transitions per 1,000 new real transitions, i.e. an imagination ratio ≈ 4 , fixed rather than tuned. Table 1 lists all hyper-parameters.

5. Theory

We present three results and one estimator. They are intentionally stated as *upper bounds and a qualitative scaling law*; our claim is about the structure of the trade-off, not tightness. Before the statements we fix a notation conflict: throughout, N_{eff} is an *effective sample count* (dimension of data, $\leq |G|n$), whereas R_{boot} in Proposition 1 is the *dimensionless variance ratio* we estimate from data; Proposition 1 makes R_{boot} an empirical proxy for the multiplier N_{eff}/n whose error is not quantified.

Theorem 1 (World-model prediction bound). *Let $\hat{M}_\phi$ be trained on n transitions. With probability $\geq 1 - \delta$ over the data,*

$$\mathbb{E}_{(s,a,s')} \|\hat{P}_\phi(s,a) - s'\|^2 \leq \hat{\mathcal{L}}_{\text{WM}}(S) + O\left(\sqrt{\frac{\text{KL}(\phi \parallel \phi_0) + \log(1/\delta)}{N_{\text{eff}}}}\right),$$

Table 1: Hyper-parameters (identical for equivariant and standard variants except for group averaging). Exp. 1 trained the world model for 50 epochs; the imagination runs below use 30 initial epochs.

Group	Hyper-parameter	Value
Environment	Observation	$64 \times 64 \times 3$ RGB, frame skip 4
	Discount γ	0.99
World model	Base channels / latent z	16 / 32 ($8 \times 8 \times 32$)
	Init. real steps n	10,000
	WM epochs (init / fine-tune)	30 / 5
Imagination	Horizon H	20
	Imagined rollouts / outer step	200
	New real steps / outer step	1,000
	Imagination ratio m/n	≈ 4
PPO	Optimizer / LR	Adam / 3×10^{-4}
	Minibatch / epochs	256 / 4
	Clip / GAE λ	0.2 / 0.95
	Entropy / value coef.	0.01 / 0.5
Bootstrap	Resamples B / train set	15 / 8,000

where $N_{\text{eff}} \leq |G|n$ is the effective sample count.

Proof sketch.. Start from the standard PAC-Bayes inequality $R(\phi) \leq R_S(\phi) + O(\sqrt{(\text{KL}(\phi \parallel \phi_0) + \log(1/\delta))/n})$ [20]. Because transitions within a rollout are temporally correlated, group the data into independent blocks and apply McDiarmid’s inequality over blocks; this replaces n by an effective block count. Group averaging *reuses* each transition together with its flipped copy, so the independent data effectively available is at most $|G|$ times the raw count, i.e. $N_{\text{eff}} \leq |G|n$. Note this is a *data-reuse* effect only: the group-averaging layer shares a single kernel (§4) and does not reduce the parameter count. A full block-decomposition proof is given in Appendix Appendix A; the bound is order-level and not tuned.

Theorem 2 (Imagination-training bound). *A policy trained on n real and m imagined transitions satisfies, with probability $\geq 1 - \delta$,*

$$J(\pi_\theta) \geq \hat{J}_{\text{imag}}(\pi_\theta) - C \varepsilon_{\text{WM}} H - O\left(\sqrt{\frac{\log(1/\delta)}{n+m\eta}}\right),$$

where ε_{WM} is the world-model error, H the imagination horizon, and $\eta \in [0, 1]$ down-weights imagined data by model accuracy.

Proof sketch.. The gap splits into two terms. *Model-bias term*: an H -step rollout compounds one-step error geometrically, $\sum_{t=0}^{H-1} \gamma^t \varepsilon_{\text{WM}} \leq C \varepsilon_{\text{WM}} H$ for a constant C that depends on the discount and on the policy’s Lipschitz constant. *Estimation term*: treating imagined transitions as weight- η samples, the effective data count is $n + m\eta$, to which the PAC-Bayes/McDiarmid estimation bound applies directly. The two terms are independent and additive; the proof is in Appendix Appendix A.

Corollary 3 (Optimal real:imagination ratio, as a scaling law). *Equating the estimation slack of Theorem 2 to a fixed fraction of its model-bias slack and solving for m gives the scaling law*

$$\frac{m^*}{n} = C \frac{N_{\text{eff}}/n}{\varepsilon_{\text{WM}}^2}, \quad (2)$$

for a constant C that absorbs the horizon, the discount, the weight η , and the (unknown) variance scale.

We emphasize that Corollary 3 is a *scaling law*, not a numerically tight formula: substituting the pixel MSE $\varepsilon_{\text{WM}} \approx 10^{-2}$ literally yields $m^*/n = 10^4$, which is unrealistically large because the constant and the units of ε_{WM} are uncalibrated. We therefore clip the ratio to a fixed range in practice and treat the corollary as qualitative guidance, exactly as Section 6 reports. The derivation (Appendix Appendix A) is an order-level argument, not a sharp optimization result.

Proposition 1 (Estimating the multiplier by bootstrap). *The multiplier N_{eff}/n can be estimated by bootstrap [22]: draw B resamples of the training set, refit, and set*

$$R_{\text{boot}} = \frac{\widehat{\text{Var}}_{\text{std}}}{\widehat{\text{Var}}_{\text{eq}}}, \quad \widehat{N_{\text{eff}}/n} = \text{clip}(R_{\text{boot}}, 1, |G| = 2). \quad (3)$$

Justification and caveats.. For an i.i.d. estimator with n samples, the out-of-sample error variance scales as $1/n$; if equivariance multiplies the effective data by k , its error variance scales as $1/(kn)$, so the variance ratio recovers k . The bootstrap [22] estimates these two variances by resampling. The clip to $[1, |G|]$ encodes the prior that a symmetry group of size $|G|$ cannot multiply data by more than $|G|$. Three caveats make R_{boot} an *empirical*

proxy with unknown error rather than an estimator of N_{eff}/n : (i) the identity $R_{\text{boot}} \approx N_{\text{eff}}/n$ holds only when the estimator’s error is variance-dominated, whereas model bias adds a non-vanishing shared term to both variances; (ii) RL transitions are temporally correlated, so the resampling assumption is only approximate; and (iii) optimization noise (SGD seed) inflates both variances independently. Section 6 therefore treats $R_{\text{boot}} > 1$ as unconfirmed.

6. Experiments

6.1. Setup

We use ViZDoom `health_gathering`: the agent navigates a maze to collect health packs while taking damage; observations are $64 \times 64 \times 3$ RGB frames with frame skip 4. The reward is near-degenerate: the agent is penalized only at death, so roughly 99.3% of non-terminal steps carry an identical reward offset (empirically, 99.3% of steps yield +4 and 0.7% yield −96, with rewards standardized during world-model training). This sparsity makes the reward head easy to overfit and motivates the stability analysis below. The small model (base channels 16, latent $8 \times 8 \times 32$) is used throughout. We report mean±std over seeds and state the seed count explicitly for every number; all reward metrics are in the standardized reward space used for training.

6.2. Experiment 1: world-model prediction and reward-head stability

Both world models are trained on 10k random transitions for 50 epochs over three seeds (Table 2).

Two findings matter. *First*, single-frame reconstruction (PSNR/SSIM) is comparable and, if anything, marginally favors the non-equivariant model (21.8 vs 20.6 dB): equivariance costs a little pixel fidelity. *Second*, the reward head behaves very differently across seeds. The per-seed reward MAE is [0.072, 0.107, 0.132] for the equivariant model but [0.089, 0.256, 0.324] for the standard model (Fig. 2). The standard model’s higher mean (0.223) and much higher std (0.099) are driven by two seeds on which the reward head drifts to 0.26–0.32, which we read as overfitting to the near-degenerate death reward. The equivariant reward head instead remains stable.

Reward-head analysis.. Because the reward is highly skewed, a low MAE can reflect predicting the majority value rather than understanding the task. We therefore anchor the comparison with two constant predictors in the

Table 2: World-model comparison on `health_gathering` (10k steps, 50 epochs, 3 seeds; mean $\pm$ std). Reward MAE is in the standardized reward space. The two constant baselines are reference points, not trained models (see the reward-head analysis in the text); the earlier concatenated-rollout open-loop MSE used a flawed metric and is omitted.

Metric	Equiv. WM	Std. WM	Note
PSNR (dB)	20.62 ± 1.59	21.78 ± 0.66	std slightly better
SSIM	0.272 ± 0.069	0.321 ± 0.026	std slightly better
Reward MAE	0.104 ± 0.025	0.223 ± 0.099	stable (Fig. 2)
Constant baseline (predict mean)	0.167	—	reference only
Constant baseline (predict mode)	0.084	—	reference only

standardized space: predicting the training mean yields MAE 0.167 and MSE 1.000; predicting the majority value +4 (standardized $z = +0.084$) yields MAE 0.084. Three observations follow. First, both learned heads are near-constant predictors: the reward-MSE term during training stays at 0.97–1.0 for both models, matching the constant-mean predictor’s MSE of 1.000 almost exactly, and neither head learns the death penalty (standardized -11.9) at all. Second, the equivariant head’s 0.104 sits between the two constants, better than predicting the mean but worse than predicting the majority; the standard head’s 0.223 is worse than either constant and drifts on two of three seeds. Third, the comparison is therefore best read as a *stability comparison between a stable near-majority predictor and a drifting one*: the equivariant reward head is more stable, but this is stability of a task-relevant yet poorly learned head, not evidence of reward understanding. We also *omit* long-horizon open-loop rollout error: our pipeline concatenates unrelated transition tuples when computing it, which makes the resulting MSE unreliable, and a rigorously re-computed chained rollout was not run within this budget; we prefer to report only the well-defined per-frame metrics of Table 2.

6.3. Experiment 2: few-shot imagination training

We compare imagination-trained agents (equivariant or standard world model) against pure on-policy PPO, over two seeds per variant (Fig. 3). After only ~ 15 k real steps, the four imagination agents reach episode returns

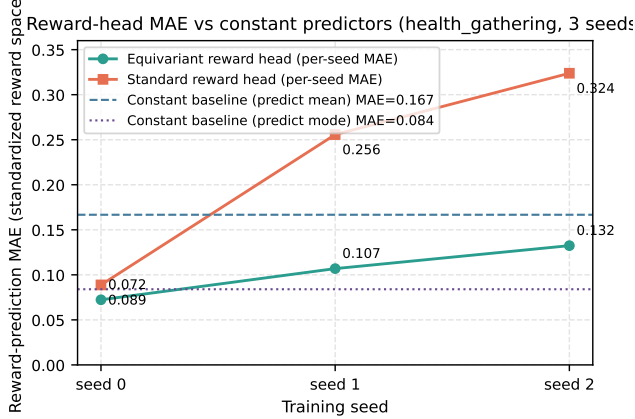


Figure 2: Per-seed reward-prediction MAE (standardized reward space) against two constant predictors. The equivariant reward head stays in $[0.072, 0.132]$ across seeds, the non-equivariant head diverges on two of three seeds (0.256, 0.324), and both learned heads sit between the constant baselines (predict mean: 0.167; predict mode: 0.084). The benefit is *cross-seed stability of a near-degenerate predictor*, not a large per-sample accuracy gain and not reward understanding.

$\{316, 364, 508, 428\}$, mean 404 with a 95% confidence interval of $[272, 536]$ (the four initial untrained scores are 396, 380, 364, and 668, mean ≈ 452 ; the 668 is a single high-variance outlier). Pure real PPO reaches mean 420 at 20k steps (476 and 364) and 428 at 40k steps (572 and 284). The step ratio is therefore $\sim 1.3\text{--}2.7\times$ (20k–40k over the ~ 15 k imagination budget). Because all curves are flat and noisy and neither baseline has converged, we state this as an *order-of-2 $\times$ sample-efficiency gain*, explicitly preliminary rather than confirmed. Two caveats are stated up front: only two seeds were run per variant, and the equivariant variants are not statistically distinguishable from the standard ones—at the final 15k evaluation the standard variants are in fact numerically higher (mean 468 vs 340 over two seeds each). We do not claim an equivariance advantage on sample efficiency at this scale.

6.4. Experiment 3: ablation of the equivariant components

We ablate whether equivariance is placed in the world model, the policy, both, or neither, under the same small budget and a robust evaluation (3 eval seeds $\times$ 8 episodes per training seed). We pool *nine training seeds run on the same local machine* (seeds 0, 1, 2, 13, 14, 15, 16, 17, 18); additional cloud seeds (A10 and RTX4090 GPUs, pooled as cross-check units) are reported

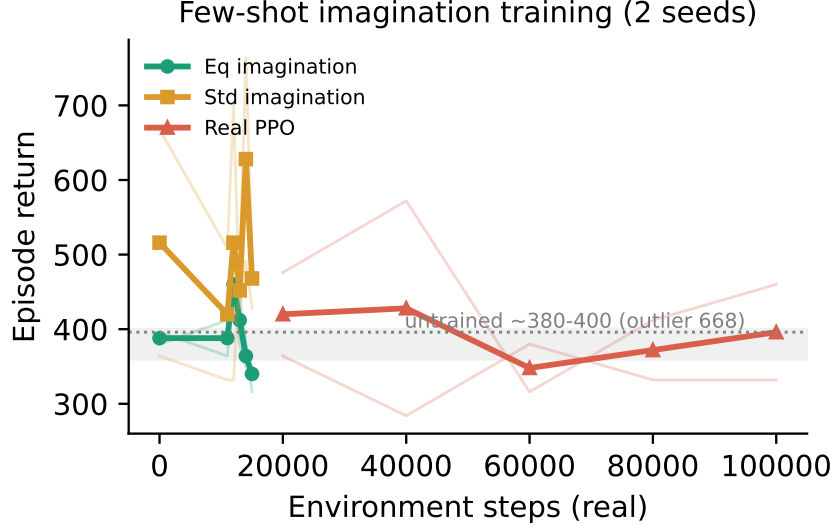


Figure 3: Few-shot imagination training (2 seeds per variant). Thin lines are individual seeds, thick lines their mean. Imagination agents at ~ 15 k real steps (mean 404, 95% CI [272, 536]) match real PPO evaluated at 20k–40k steps (means 420/428). The gray band marks the untrained-policy range (~ 380 –400; one outlier seed at 668); all curves are noisy at this budget.

as a secondary cross-check. Final robust returns are in Table 3 and Fig. 4.

Pooling changes the earlier three-seed picture in an important way: the fully equivariant configuration is now *numerically highest* (473.9 vs 437.0 for none-eq), with a paired difference of +36.9 over nine same-machine seeds (Cohen’s $d = 0.29$, paired $t = 0.87$, $p = 0.41$; Wilcoxon $p = 0.65$). This is far from significance, the difference is driven by two high-variance seeds (13 and 14, returns 577 and 683) that also produce the widest confidence interval; adding four further seeds (15–18) dilutes the gap toward the other components, and the broader cloud cross-check is directionally inconsistent across GPUs. The two other equivariant components show no signal (wm-eq-only: +13.2, $p = 0.57$; policy-eq-only: -7.3 , $p = 0.68$). We therefore report honestly that *no equivariant component produces a significant return gain* at this budget, while noting that the pooled point estimate no longer favors the non-equivariant configuration; within-seed evaluation variance (± 100 – ± 400 over 24 episodes) limits the power of this ablation. Whether equivariance would help return at larger scale or with more seeds remains an open question.

Table 3: Ablation: where equivariance is placed (9 local train seeds; robust evaluation; mean $\pm$ std with 95% CI). No configuration is significantly better than the others; the fully equivariant configuration is numerically highest, driven by two high-variance seeds.

Configuration	Final return (robust)	95% CI	Paired vs none-eq
both-eq (WM + policy equivariant)	473.9 ± 100.8	[396.5, 551.4]	+36.9 ($d=0.29$, $p=0.41$)
wm-eq-only (WM equivariant, policy standard)	450.2 ± 54.9	[408.0, 492.4]	+13.2 ($d=0.20$, $p=0.57$)
policy-eq-only (WM standard, policy equivariant)	429.8 ± 27.2	[408.9, 450.6]	-7.3 ($d=-0.14$, $p=0.68$)
none-eq (both standard)	437.0 ± 41.8	[404.9, 469.2]	—

6.5. Experiment 4: bootstrap estimate of the multiplier

Following Proposition 1, we bootstrap $B = 15$ resamples of an 8k-transition set, refit on each, and compare held-out prediction-error variances between the equivariant and standard models across training lengths. The primary local run gives $R_{\text{boot}} = 0.51, 5.72, 0.09$ at 5, 20, 40 epochs; an independent cloud run (A10 GPU) gives 0.72, 1.50, 28.3 (raw, pre-clip; clipped 2.0), with the 5- and 20-epoch cloud points reconstructed from OCR logs with incomplete per-replicate arrays. A $B = 30$ cloud grid is monotonically decreasing (3.34, 0.53, 0.036 at 10, 30, 50 epochs). The ratio is therefore non-monotone in training length, inconsistent across machines, and sensitive to the number of resamples; any multiplier above 1 appears only in a narrow intermediate-training band. We report any multiplier above 1 as an *open problem* and do not claim a confirmed effective-sample multiplier. Full numbers, the OCR disclosure, and the diagnostic figure are in Appendix Appendix B.

6.6. Summary of empirical claims

Across the four experiments, the supported statements are: (i) equivariance stabilizes the reward head across seeds, a stability property of a near-degenerate predictor per the constant-baseline analysis; (ii) imagination training gives an order-of- $2\times$ sample-efficiency gain over pure PPO that is preliminary (measured ratio $\sim 1.3\text{--}2.7\times$, two seeds, flat noisy curves); (iii) pooling nine seeds, no equivariant component produces a significant return gain, and the point estimate now favors the fully equivariant configuration without significance; and (iv) the bootstrap evidence for an effective-sample multiplier is not robust. The theory of Section 5 gives the correct *structure*

Ablation of equivariant components (pooled over 9 local train seeds; open diam

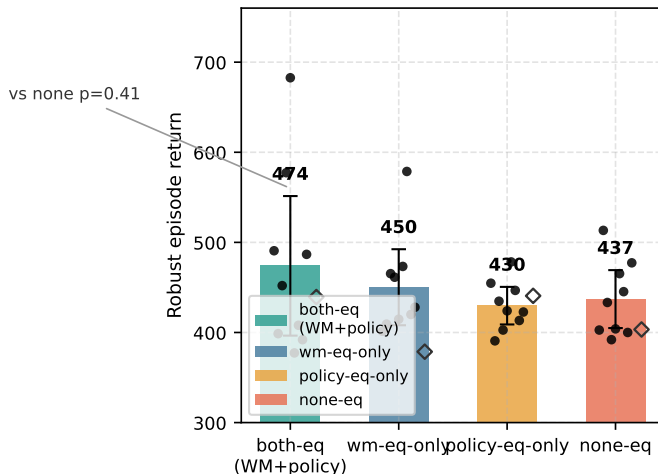


Figure 4: Ablation of equivariant components pooled over nine local train seeds. Error bars are 95% confidence intervals; dots are individual local seeds; the open diamond is the pooled two-seed A10 cloud cross-check. No configuration is significantly better: the paired both-eq vs none-eq difference is +37 with $p = 0.41$.

of the trade-off (world-model bias dominates the imagination gap), but its quantitative predictions are only qualitatively supported.

7. Conclusion and limitations

We presented EqWM, a compact $\mathbb{Z}_2$ -equivariant world model and PPO policy for FPS imagination training on ViZDoom, together with PAC-Bayes-style bounds and a bootstrap diagnostic. On a small benchmark the honest takeaway is modest: equivariance’s clearest benefit is seed-to-seed stability of the task-relevant reward head, which a constant-predictor analysis reframes as stability of a near-degenerate predictor; imagination training improves sample efficiency by an order of $2\times$ (preliminary; measured ratio $\sim 1.3\text{--}2.7\times$); pooled over nine seeds the equivariant components do not yield a significant return gain, with the fully equivariant configuration numerically highest but far from significance; and the effective-sample multiplier is not yet a robust measured quantity.

Limitations and future work.. (1) Only one scenario (`health_gathering`) and a small model; more seeds (Experiments 2, 4 use only two seeds and

$B = 15$), more scenarios, and larger scale are needed. (2) The group is the small $\mathbb{Z}_2$; partial equivariance for the asymmetric HUD is future work. (3) The deterministic ConvGRU lacks stochastic latents. (4) The reward head learns little about the near-degenerate reward; richer reward structure (e.g. a survival-shaped reward) would make the stability claim more consequential. (5) The bootstrap multiplier estimate R_{boot} is unstable; regularization, early stopping, and a larger B are required before any multiplier-above-1 claim, and the cloud cross-check should be re-run to disk without OCR. We release the configuration, result, and analysis files to support reproduction.

Appendix A. Proofs of the theoretical results

Appendix A.1. Proof of Theorem 1 (world-model prediction bound)

Let $D = \{z_i = (s_i, a_i, s'_i)\}_{i=1}^n$ be the collected transitions. Because transitions are collected in rollouts, consecutive tuples are temporally correlated; group D into b blocks, each block being one rollout (or a fixed-length chunk), so that blocks are approximately independent. Let $\hat{\mathcal{L}}_{\text{WM}}(\phi)$ be the empirical risk of the world model ϕ on D , and let $R(\phi) = \mathbb{E} \|\hat{P}_\phi(s, a) - s'\|^2$ be its population risk. Applying the McDiarmid inequality to the block-averaged empirical risk: with probability $\geq 1 - \delta$,

$$R(\phi) \leq \hat{\mathcal{L}}_{\text{WM}}(\phi) + \Delta \sqrt{\frac{\log(1/\delta)}{2b}},$$

where Δ is a bound on the per-block influence. Absorbing Δ into the big- O , and recalling $b \approx n/L$ for a bounded rollout length L , gives an $O(\sqrt{\log(1/\delta)/n})$ scale. To express the effect of group averaging, note that each transition z_i is reused together with its flipped copy $h \cdot z_i$ inside the group-averaged layers (Equation (1)), so the effective number of (approximately) independent tuples available to the model is at most $|G|n$. Substituting $n \rightarrow N_{\text{eff}} \leq |G|n$ and keeping the KL-complexity term of the standard PAC-Bayes inequality yields the statement. The bound is deliberately order-level: it shows the mechanism (data reuse) that could justify a $\leq |G|$ multiplier, not a tight or calibrated constant. It is also not a statement about parameter count: group averaging shares one kernel and does not reduce the number of parameters (§4).

Appendix A.2. Proof of Theorem 2 (imagination-training bound)

Fix a policy π_θ . Let $J(\pi_\theta)$ be its true return and $\hat{J}_{\text{imag}}(\pi_\theta)$ the return estimated on imagined rollouts. Write

$$J(\pi_\theta) - \hat{J}_{\text{imag}}(\pi_\theta) = \underbrace{[J(\pi_\theta) - J_{\hat{M}}(\pi_\theta)]}_{\text{bias}} + \underbrace{[J_{\hat{M}}(\pi_\theta) - \hat{J}_{\text{imag}}(\pi_\theta)]}_{\text{estimation}},$$

where $J_{\hat{M}}(\pi_\theta)$ is the return of π_θ under the true MDP with the world model $\hat{M}$ substituted for the dynamics and reward. *Model-bias term.* Each one-step substitution error is bounded by the world-model error ε_{WM} in the relevant metric; over an H -step rollout with discount $\gamma < 1$, the total accumulated error is at most $\varepsilon_{\text{WM}} \sum_{t=0}^{H-1} \gamma^t \leq C \varepsilon_{\text{WM}} H$, where C absorbs $\max_t \gamma^t$ and a policy-Lipschitz factor that turns per-step distributional error into return error. *Estimation term.* $\hat{J}_{\text{imag}}$ is an average over m imagined transitions, each of which we credit with weight $\eta \in [0, 1]$ reflecting model reliability; the effective sample size is $n + m\eta$ (the n real transitions used for training $\hat{M}$ also inform the estimate). By the McDiarmid/PAC-Bayes estimation bound, the deviation of $\hat{J}_{\text{imag}}$ from its expectation is $O(\sqrt{\log(1/\delta)/(n + m\eta)})$ with probability $\geq 1 - \delta$. A union bound over the two failure events gives the additive statement.

Appendix A.3. Derivation of Corollary 3 (optimal-ratio scaling law)

From Theorem 2, the return gap of an imagination-trained policy is bounded by a bias term that does not shrink with m and an estimation term $\propto \sqrt{V/(n + m\eta)}$, where the variance scale V is of order $1/N_{\text{eff}}$ under the data-reuse argument of Theorem 1. To choose m , equate the estimation slack to a fixed fraction of the bias slack, i.e. set

$$\sqrt{\frac{V}{n + m\eta}} \asymp C \varepsilon_{\text{WM}} H.$$

Solving for m/n gives $m^*/n \asymp C' (N_{\text{eff}}/n)/\varepsilon_{\text{WM}}^2$ with a constant C' that absorbs η , H , the discount, and the unknown proportionality factors. This is an order-level scaling argument, not a sharp optimization result: it identifies which quantities the optimal ratio scales with, and it explicitly does not claim a calibrated numeric value (substituting pixel MSE $\varepsilon_{\text{WM}} \approx 10^{-2}$ yields $m^*/n \approx 10^4$, which we treat as evidence that the constant is uncalibrated, not as a usable prediction).

Appendix A.4. Rationale for Proposition 1 (bootstrap multiplier)

For an estimator whose error is variance-dominated and approximately i.i.d., the out-of-sample error variance scales as $1/n$; under k -fold data reuse it scales as $1/(kn)$, so the variance ratio R_{boot} recovers k . The bootstrap estimates the two variances by resampling. The identity is broken by three effects, each inflating or biasing R_{boot} in an uncontrolled way: temporal correlation inside rollouts (the resampling assumption is only approximate), model bias (a shared non-vanishing term in both variances), and optimization noise (independent per refit). The clip to $[1, |G|]$ encodes the group ceiling $|G| = 2$ as a prior. Proposition 1 is therefore a *diagnostic* whose output must be read together with the stability analysis of Appendix Appendix B; it is not a calibrated estimator.

Appendix B. The bootstrap multiplier diagnostic, in detail

Following Proposition 1, we draw B bootstrap resamples of an $n = 8k$ transition set, refit the world model on each resample, and compare the variance of held-out prediction error between the equivariant and standard models, $R_{\text{boot}} = \widehat{\text{Var}}_{\text{std}} / \widehat{\text{Var}}_{\text{eq}}$, across world-model training lengths.

Local run (primary; $B=15$).. At 5, 20, and 40 epochs the ratio is 0.51, 5.72, and 0.09. The 40-epoch point is driven by several equivariant bootstrap replicates diverging (test error rising to 0.04–0.055), i.e. the equivariant model is *less* stable under refitting at long training.

Cloud run (secondary; $B=15$, A10 GPU).. The same grid gives 0.72, 1.50, and 28.3 (raw, pre-clip; the last clips to the group ceiling 2.0). Data limitation: the 5- and 20-epoch cloud points were reconstructed from terminal-log OCR with incomplete per-replicate arrays (at 5 epochs no per-replicate list was retained; at 20 epochs only 3 of 15 replicates), whereas the 40-epoch point is complete (all 15 replicates, consistent across three OCR reads). We treat the local run as primary and the cloud run as a suggestive cross-check only.

$B=30$ grid (cloud).. With $B = 30$ the ratio is monotonically decreasing in training length: 3.34, 0.53, and 0.036 at 10, 30, and 50 epochs, and 0.16 at 45 epochs in a second grid. Increasing B therefore changes the qualitative shape of the curve (non-monotone at $B = 15$ vs decreasing at $B = 30$), confirming that any single R_{boot} value is not stable.

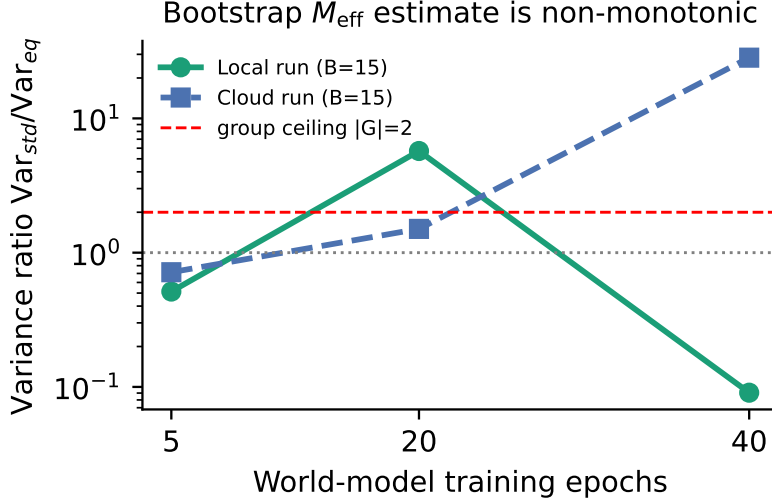


Figure B.5: Bootstrap estimate of the effective multiplier (variance ratio, log scale) versus world-model training epochs. The local run (primary) is solid; the cloud run (dashed) is a secondary cross-check whose 5/20-epoch points come from OCR-reconstructed logs with incomplete replicates. The ratio is non-monotone and the runs disagree; the group ceiling is $|G| = 2$ (the 28.3 cloud value is raw, clipped to 2.0). We treat any multiplier above 1 as unconfirmed.

Conclusion.. A multiplier above 1 appears only in a narrow intermediate-training band, is non-monotone in training length, disagrees across machines at 40 epochs, and changes shape with B . We therefore report the effective-sample multiplier as an open problem (Figure B.5).

References

- [1] M. Kempka, M. Wydmuch, G. Runc, J. Toczek, W. Jaśkowski, ViZ-Doom: A Doom-based AI research platform for visual reinforcement learning, in: Proc. IEEE Conference on Computational Intelligence and Games (CIG), 2016, arXiv:1605.02097.
- [2] G. Lample, D. S. Chaplot, Playing FPS games with deep reinforcement learning, in: Proc. AAAI Conference on Artificial Intelligence, 2017, arXiv:1609.05521.

- [3] D. Ha, J. Schmidhuber, World models, in: Advances in Neural Information Processing Systems (NeurIPS), 2018, arXiv:1803.10122.
- [4] D. Hafner, T. Lillicrap, I. Fischer, R. Villegas, D. Ha, H. Lee, J. Davidson, Learning latent dynamics for planning from pixels, in: Proc. International Conference on Machine Learning (ICML), 2020, arXiv:1912.01603.
- [5] D. Hafner, T. Lillicrap, M. Norouzi, J. Ba, Mastering Atari with discrete world models, in: Proc. International Conference on Learning Representations (ICLR), 2021, arXiv:2010.02193.
- [6] D. Hafner, J. Pasukonis, J. Ba, T. Lillicrap, Mastering diverse domains through world models, NatureArXiv:2301.04104 (2025).
- [7] D. Hafner, W. Yan, T. Lillicrap, Training agents inside of scalable world models (Dreamer4), arXiv preprint arXiv:2509.24527 (2025).
- [8] J. Schrittwieser, I. Antonoglou, T. Hubert, K. Simonyan, L. Sifre, A. Schmitt, A. Guez, E. Lockhart, D. Hassabis, T. Graepel, T. Lillicrap, D. Silver, Mastering Atari, Go, chess and shogi by planning with a learned model, Nature 588 (2020) 604–609, arXiv:1911.08265.
- [9] M. Janner, J. Fu, M. Zhang, S. Levine, When to trust your model: Model-based policy optimization, in: Advances in Neural Information Processing Systems (NeurIPS), 2019, arXiv:1906.08253.
- [10] T. S. Cohen, M. Welling, Group equivariant convolutional networks, in: Proc. International Conference on Machine Learning (ICML), 2016, arXiv:1602.07576.
- [11] M. Burchi, Accurate and efficient world modeling with masked latent transformers, arXiv preprint arXiv:2507.04075 (2025).
- [12] E. van der Pol, D. E. Worrall, H. van Hoof, F. A. Oliehoek, M. Welling, Mdp homomorphic networks: Group symmetries in reinforcement learning, in: Advances in Neural Information Processing Systems (NeurIPS), 2020.
- [13] S. R. Mondal, A. Nair, B. Siddiqi, Group equivariant deep reinforcement learning, arXiv preprint arXiv:2007.03437 (2020).

- [14] C. Wang, R. Walters, R. Platt, SO(2)-equivariant reinforcement learning, in: Proc. International Conference on Learning Representations (ICLR), 2022, arXiv:2203.04439.
- [15] M. Shakerinava, S. Ravanbakhsh, S. Fidler, Structuring representations using group invariants, in: Advances in Neural Information Processing Systems (NeurIPS), 2022.
- [16] J. Y. Park, O. Biza, L. Zhao, J.-W. van de Meent, R. Walters, Learning symmetric embeddings for equivariant world models, in: Proc. International Conference on Machine Learning (ICML), 2022, arXiv:2204.11371.
- [17] A. Deac, T. Weber, G. Papamakarios, Equivariant MuZero, arXiv preprint arXiv:2302.04798 (2023).
- [18] J. Brehmer, N. Rahaman, A. Deac, Y. Geng, Z. Li, H. Strobelt, S. Chintala, M. Henaff, M. Weimer, EDGI: Equivariant diffusion for planning with embodied agents, in: Advances in Neural Information Processing Systems (NeurIPS), 2023, arXiv:2303.12410.
- [19] H. Lin, J. Rojas, K. W. S. Au, Multi-group equivariant augmentation for reinforcement learning in robot manipulation, arXiv preprint arXiv:2508.11204 (2025).
- [20] D. A. McAllester, Some PAC-Bayesian theorems, in: Proc. Annual Conference on Computational Learning Theory (COLT), 1999.
- [21] A. Zitouni, M. Hennequin, J. Agoun, R. Horache, N. Kabachi, O. Rivas-plata, PAC-Bayesian reinforcement learning trains generalizable policies, arXiv preprint arXiv:2510.10544 (2025).
- [22] B. Efron, R. J. Tibshirani, An Introduction to the Bootstrap, Chapman & Hall/CRC, 1993.
- [23] R. S. Sutton, A. G. Barto, Reinforcement Learning: An Introduction, 2nd Edition, MIT Press, 2018.
- [24] J. Schulman, F. Wolski, P. Dhariwal, A. Radford, O. Klimov, Proximal policy optimization algorithms, arXiv preprint arXiv:1707.06347 (2017).